



Implementation of multiple control methods on a 2-DOF Helicopter

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- 1 Motivation
- 2 Problem formulation
- 3 (Optimisation-based) Control Strategies
 - Pole Placement
 - Linear Quadratic Regulator
 - Linear Quadratic Integrator
 - Model-based Predictive Controller
- 4 Implementation
 - Pole Placement
 - Linear Quadratic Regulator
 - Linear Quadratic Integrator
 - MPC - target setpoint
 - MPC - reference tracking
- 5 Conclusions and future work

Outline

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Motivation



Quanser Aero 2 plant

Why Quanser Aero 2?

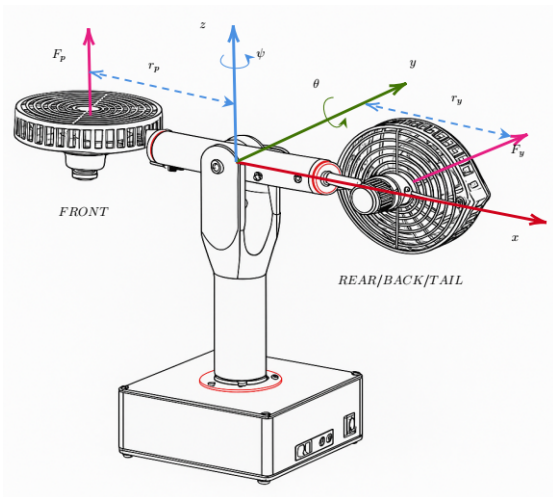
A safe laboratory testbed for UAV attitude control, with nonlinear, coupled pitch–yaw dynamics.

- Reproduces key rotorcraft challenges: nonlinearity and pitch–yaw coupling
- Safe, repeatable alternative to flight testing
- Reconfigurable (1-DOF, 2-DOF helicopter, half-quadrotor)
- interaction with both Simulink and embedded system

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Dynamical Model



- state $x = [\theta \quad \psi \quad \dot{\theta} \quad \dot{\psi}]^T \in \mathbb{R}^4$:
 - θ [rad] - pitch angular position
 - ψ [rad] - yaw angular position
 - $\dot{\theta}$ [rad/s] - pitch angular speed
 - $\dot{\psi}$ [rad/s] - yaw angular speed
- input $u = [V_p \quad V_y]^T \in \mathbb{R}^2$
 - V_p [V] - voltage input of the front motor
 - V_y [V] - voltage input of the back motor
- constraints:
 - hard: $V_p, V_y \in [-24 \text{ V}, +24 \text{ V}]$
 - soft: $\theta \in [-0.89 \text{ V}, +0.89 \text{ rad}]$

Nonlinear model and linearization

The nonlinear model $\dot{x} = f(x, u)$ is defined by the dynamics around the pitch (θ) and yaw (ψ) axes

$$\begin{aligned}(J_p + ml^2) \ddot{\theta} &= D_t K_{pp} V_p + D_t K_{py} V_y - D_p \dot{\theta} - K_{sp} \theta - mgl \cos(\theta) - m\dot{\psi}^2 l^2 \sin(\theta) \cos(\theta) \\ (J_y + ml^2 \cos^2(\theta)) \ddot{\psi} &= D_t K_{yp} V_p + D_t K_{yy} V_y - D_y \dot{\psi} + ml^2 \dot{\psi} \dot{\theta} \sin(2\theta)\end{aligned}$$

Modifications from^a

- $K_{sp}\theta$ - for symmetry - added for system **stabilization**
- D_t [m] - the distance between the yaw axis and any of the motors
 - added for dimensionality reasons, K_{pp} being measured in [N/V]

Linearization to: $\dot{x} = Ax + Bu$

- ① find the equilibrium: (x^*, u^*) s.t. $f(x^*, u^*) = 0$
- ② Taylor approximation:

$$f(x, u) \approx \left. \frac{\partial f}{\partial x} \right|_{(x^*, u^*)} (x - x^*) + \left. \frac{\partial f}{\partial u} \right|_{(x^*, u^*)} (u - u^*)$$

- ③ identify:

$$A = \left. \frac{\partial f}{\partial x} \right|_{(x^*, u^*)}, \quad B = \left. \frac{\partial f}{\partial u} \right|_{(x^*, u^*)}$$

^aLambert et al. 2018.

Problem formulation - Objectives

Purpose: build a controller that designs the command input $u = g(x)$ based on the linear approximation

- stabilize the system
- track any reference signal for the pitch θ and yaw ψ positions

Why stabilize?

$\frac{\partial \ddot{\psi}}{\partial \psi} = 0$, but $\frac{\partial \ddot{\psi}}{\partial \dot{\psi}} \neq 0 \implies$ the yaw dynamics are marginally stable, as it has an integrator in its transfer functions

Proposed solutions based on the linear approximation:

- 1 Pole-Placement controller - **PP**
- 2 Linear Quadratic Regulator - **LQR**
- 3 Linear Quadratic Integrator - **LQI**
- 4 Model-based Predictive Controller - **MPC**

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Pole Placement

Pole Placement

If the system (A, B) is controllable, one can impose a stable desired spectrum $\Lambda_d \subset \mathbb{C}_-$ by computing the gain $K \in \mathbb{R}^{m \times n}$. The gain determines the command

$$u = -Kx \implies \dot{x} = (A - BK)x$$

and closes the loop in the desired spectrum $\Lambda(A - BK) = \Lambda_d$.

Advantages:

- simple implementation
- offline calculation

Disadvantages:

- **unintuitive impact on the state and command**
- **no constraint handling** $\implies$ the inputs have to be projected into the constraint domain

Linear Quadratic Regulator

Linear Quadratic Regulator - target setpoint

We prioritize the low command u and a low steady-state error $x - x_{target}$ by choosing penalizing matrices $Q \in \mathbb{R}^{4 \times 4}$ and $R \in \mathbb{R}^{2 \times 2}$ and solve the optimization problem:

$$\min_{u \in \mathbb{R}^2} \int_0^{+\infty} \left[(x - x_{target})^T Q (x - x_{target}) + u^T R u \right] dt$$

$$\text{s.t.: } \dot{x} = Ax + Bu$$

and compute the gain K for which $u_o = -Kx$

Advantages:

- simple implementation
- intuitive impact on the state and command (e.g.: the bigger $\|R\|$, the smaller the command input)
- closed-form solution: $K = -R^{-1}B^T P$ via the **Riccati** equation

Disadvantages:

- **no constraint handling**
- the output may have a steady-state error

Linear Quadratic Integrator

Linear Quadratic Integrator

- state augmentation

$$x_a = \left[\int \theta \quad \int \psi \quad \theta \quad \psi \quad \dot{\theta} \quad \dot{\psi} \right]^T \in \mathbb{R}^6$$

- system augmentation: $\dot{x}_a = A_a x_a + B_a u$
- solve the LQR problem with a desired state x_a^{target} and input u_{target}

$$\min_{u \in \mathbb{R}^2} \int_0^{+\infty} \left[(x_a - x_a^{target})^T Q_a (x_a - x_a^{target}) + (u - u_{target})^T R (u - u_{target}) \right] dt$$

$$\text{s.t.: } \dot{x}_a = A_a x_a + B_a u, \quad \dot{x}_a^{target} = A_a x_a^{target} + B_a u_{target}$$

Advantages:

- same as the LQR
- ensured no steady-state error

Disadvantages:

- no constraint handling**
- ~~the output may have a steady-state error~~

MPC - target setpoint

Target setpoint problem

$$\min_{\substack{i \in \overline{0, N_{\text{pred}}}, \\ u_{k+i} \in \mathbb{R}^2}} \sum_{i=0}^{N_{\text{pred}}} \left[(x_{k+i} - x_{\text{target}})^\top Q (x_{k+i} - x_{\text{target}}) + u_{k+i}^\top R u_{k+i} \right] + (x_{k+N_{\text{pred}}+1} - x_{\text{target}})^\top P (x_{k+N_{\text{pred}}+1} - x_{\text{target}})$$

$$\text{s.t.: } x_{k+i+1} = A_d x_{k+i} + B_d u_{k+i}, \quad i \in \overline{0, N_{\text{pred}}}$$

$$x_{k+i} \in \mathcal{X}, \quad i \in \overline{0, N_{\text{pred}} + 1}$$

$$u_{k+i} \in \mathcal{U}, \quad i \in \overline{0, N_{\text{pred}}}$$

$$\Delta u_i = u_{k+i} - u_{k+i-1} \in \tilde{\mathcal{U}}, \quad i \in \overline{0, N_{\text{pred}}}, \quad u_{-1} = u_{\text{init}}$$

N_{pred} - prediction horizon

Advantages:

- **constraint handling**
- intuitive impact on the state and command
- possible preemptive behaviour

Disadvantages:

- ~~no constraint handling~~
- online computation
- loss of guarantee of closed-loop stability

MPC - reference tracking

Reference tracking problem

$$\begin{aligned}
 & \min_{\substack{i \in \overline{0, N_{pred}} \\ u_{k+i} \in \mathbb{R}^2}} \sum_{i=0}^{N_{pred}} \left[(x_{k+i} - x_{k+i}^{ref})^T Q (x_{k+i} - x_{k+i}^{ref}) + \Delta u_i^T R \Delta u_i \right] + (x_{k+N_{pred}+1} - x_{k+N_{pred}+1}^{ref})^T P (x_{k+N_{pred}+1} - x_{k+N_{pred}+1}^{ref}) \\
 & \text{s.t.: } x_{k+i+1} = A_d x_{k+i} + B_d u_{k+i}, \quad i \in \overline{0, N_{pred}} \\
 & \quad x_{k+i} \in \mathcal{X}, \quad i \in \overline{0, N_{pred} + 1} \\
 & \quad u_{k+i} \in \mathcal{U}, \quad i \in \overline{0, N_{pred}} \\
 & \quad \Delta u_{k+i} = u_{k+i} - u_{k+i-1} \in \tilde{\mathcal{U}}, \quad i \in \overline{0, N_{pred}}, \quad u_{-1} = u_{init}
 \end{aligned}$$

Advantages:

- constraint handling
- intuitive impact on the state and command
- possible preemptive behaviour

Disadvantages:

- online high-complexity computation
- loss of guarantee of closed-loop stability

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Linear Model computation

Fixed the equilibrium in $x^* = \mathbf{0}_{4 \times 1}$ and obtained the dynamic matrices:

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -\frac{K_{sp}}{J_p + ml^2} & 0 & -\frac{D_p}{J_p + ml^2} & 0 \\ 0 & 0 & 0 & -\frac{D_y}{J_y + ml^2} \end{bmatrix}, B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ \frac{D_t K_{pp}}{J_p + ml^2} & \frac{D_t K_{py}}{J_p + ml^2} \\ \frac{D_t K_{yp}}{J_y + ml^2} & \frac{D_t K_{yy}}{J_y + ml^2} \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}, D = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Properties

- **marginally stable** in an open loop ^a
- **controllable**
- **observable**

^abecause of the integrator in the yaw dynamics

Next step

The described methods **can be implemented** on this linear model.

Model testing

Testing

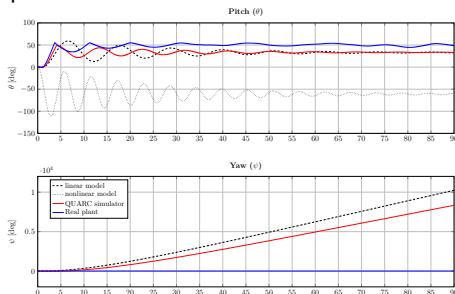
Studied the outputs of

- the nonlinear model
- the linear model
- real plant
- QUARC Simulator of the Quanser Aero 2

at identical and fixed command inputs.

Remarks

- the linear model and the QUARC simulator have similar results
- *the real plant needs bigger inputs than the simulator for similar results - Hypothesis*
- the pitch response of the nonlinear model does not fit any of the other 3 responses



Studied parameters

Reference outputs

Square wave signals:

- different amplitudes
- different frequencies

Consequence

Appearance of a *inner disturbance*, measurable by the **coupling overshoot (+)/undershoot(-)**

Pitch/Yaw output position

- Overshoot - σ [%]
- Peak Time - t_p [s]
- Steady-state error - ϵ [%]
- Coupling overshoot - ϵ' [%]

Simulation details

- simulation environment: MATLAB/Simulink
- simulation time: 30 sec
- models tested:
 - QUARC simulator
 - real Quanser Aero 2 plant

Command inputs

- median value
- amplitude and frequency of the oscillations
- comparisons between the real plant and the simulator, to test the Hypothesis

PP controller 1/3 - approach and remarks

Approach

- Imposed identical overshoots σ_d and peak times t_{p_d} for each motion to obtain a desired spectrum:

$$\sigma = 100e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}} \Rightarrow \zeta = \frac{\ln\left(\frac{100}{\sigma}\right)}{\sqrt{\pi^2 + \ln^2\left(\frac{100}{\sigma}\right)}}$$

$$t_p = \frac{\pi}{\omega_n \sqrt{1-\zeta^2}} \Rightarrow \omega_n = \frac{\pi}{t_p \sqrt{1-\zeta^2}}$$

$$p = -\zeta\omega_n \pm j\omega_n \sqrt{1-\zeta^2}$$

- applied the `place` function in MATLAB to obtain the K gain

How do we handle the input constraints?

Applied an smooth approximation of the saturation for the command using the `tanh` function

Remarks

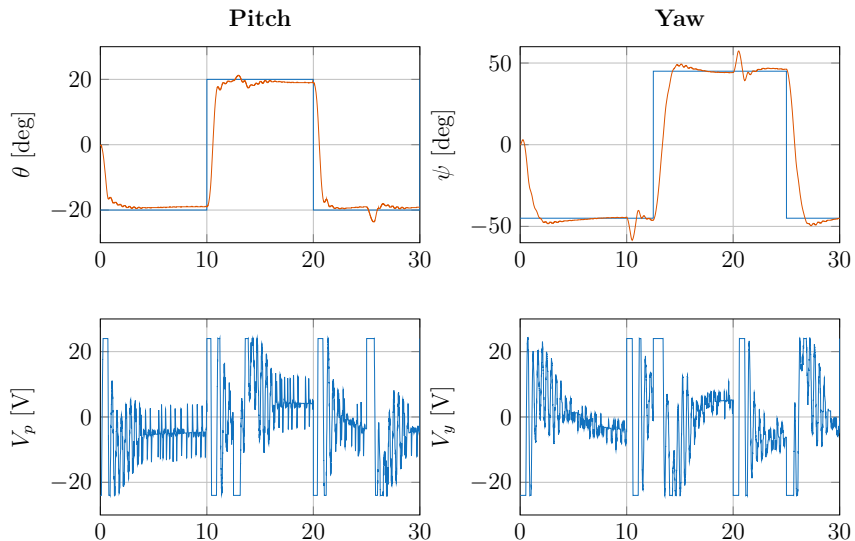
- $\sigma \geq \sigma_d$, $t_p \geq t_{p_d}$
- extremely high coupling overshoots
- while the command inputs have relatively small medium values, they have **high amplitudes and high frequencies**

Next step: LQR

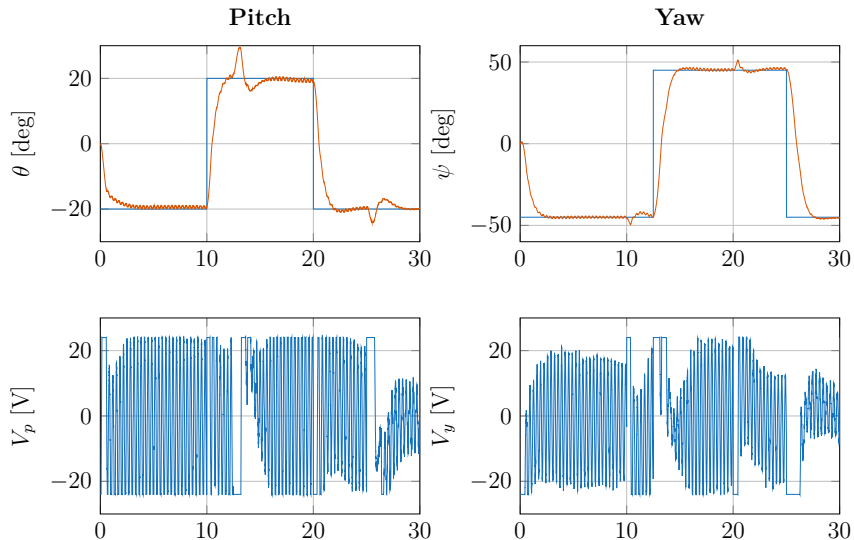
Problem: Command input with *high amplitudes and high frequencies*

Solution: a controller that reaches **high** performances and **low** commands

PP controller 2/3 - Simulator results



PP controller 3/3 - Real plant results



LQR controller 1/3 - approach and remarks

Approach

- chose the $Q = \text{diag}(10^4 \quad 10^3 \quad 10 \quad 50)$ and $R = \text{diag}(0.1 \quad 0.1) = 0.1I_2$ matrices 2 diagonal matrices:

$$\begin{bmatrix} V_p & V_y \end{bmatrix} \begin{bmatrix} 0.1 & 0 \\ 0 & 0.1 \end{bmatrix} \begin{bmatrix} V_p \\ V_y \end{bmatrix} = 0.1 V_p^2 + 0.1 V_y^2$$

- called the `lqr` function and obtained a spectrum $\Lambda \subset \mathbb{C}_-$

Remarks

- small ϵ on the pitch axis
- command inputs with oscillations with smaller amplitudes
- high sensitivity of the coupling reference change

How do we handle the input constraints?

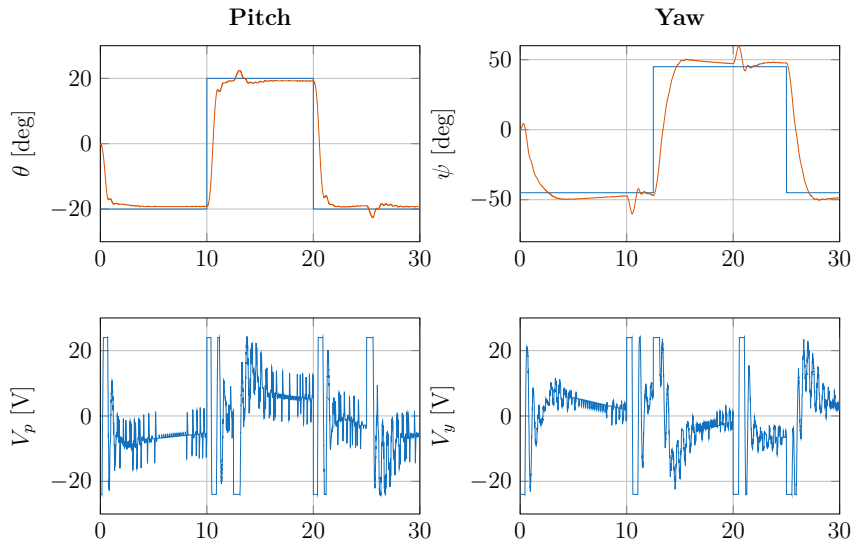
Applied an smooth approximation of the saturation for the command using the `tanh` function

Next step: LQI

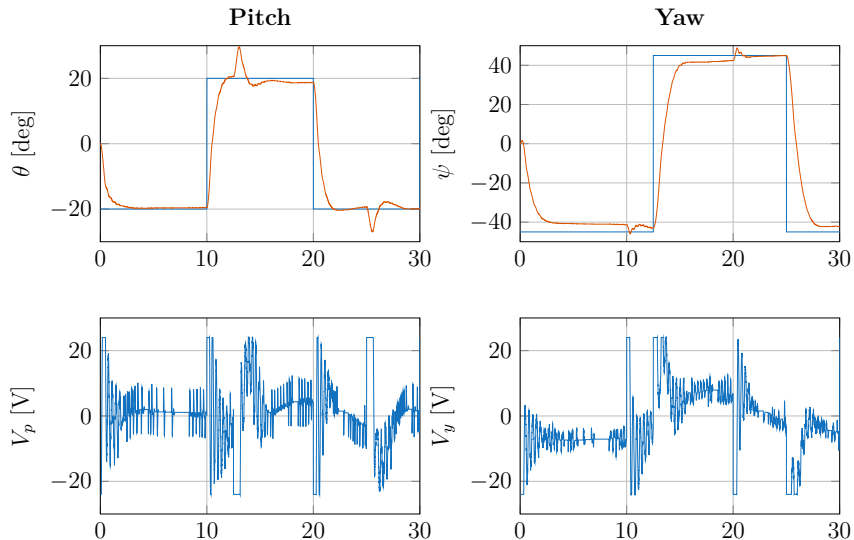
Problem: steady-state error on one of the axes

Solution: consider the state error in the optimization problem

LQR controller 2/3 - Simulator results



LQR controller 3/3 - Real plant results



LQI controller 1/3 - approach and remarks

Approach

- $Q = \text{diag}(10^{-1} \quad 10^{-2} \quad 5 \cdot 10^4 \quad 5 \cdot 10^4 \quad 2 \cdot 10^4 \quad 10^4)$
- $R = I_2$
- the desired command reference in Simulink is calculated starting from the formula

$$\dot{x}_a^{\text{target}} = A_a x_a^{\text{target}} + B_a u_{\text{target}}$$

$$\underbrace{B_a}_{A} \underbrace{u_{\text{target}}}_{x} = \underbrace{\dot{x}_a^{\text{target}} - A_a x_a^{\text{target}}}_{y}$$

- the obtained spectrum is stable, but not robust stable

How do we handle the input constraints?

Applied an smooth approximation of the saturation for the command using the [tanh](#) function

Remarks

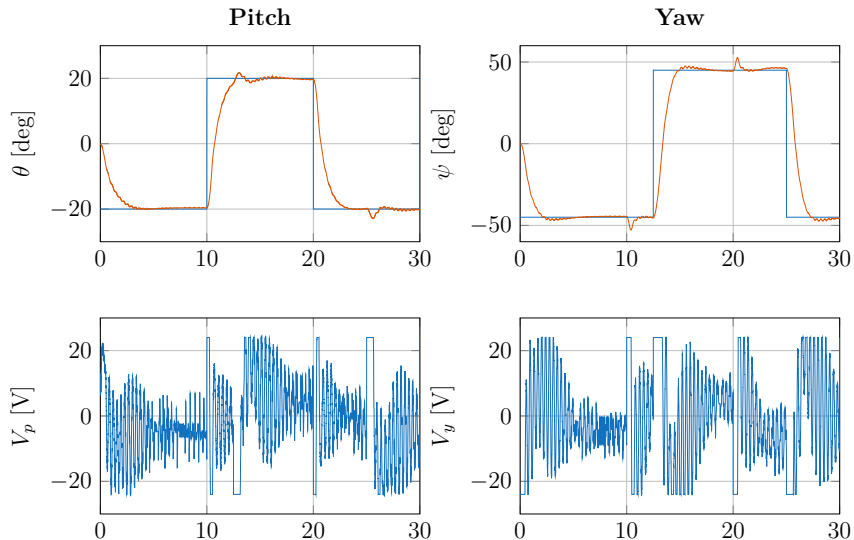
- slower settling time
- $\epsilon = 0\%$
- commands comparable with the PP controller
- extremely high sensitivity in high coupling overshoots

Next step: MPC

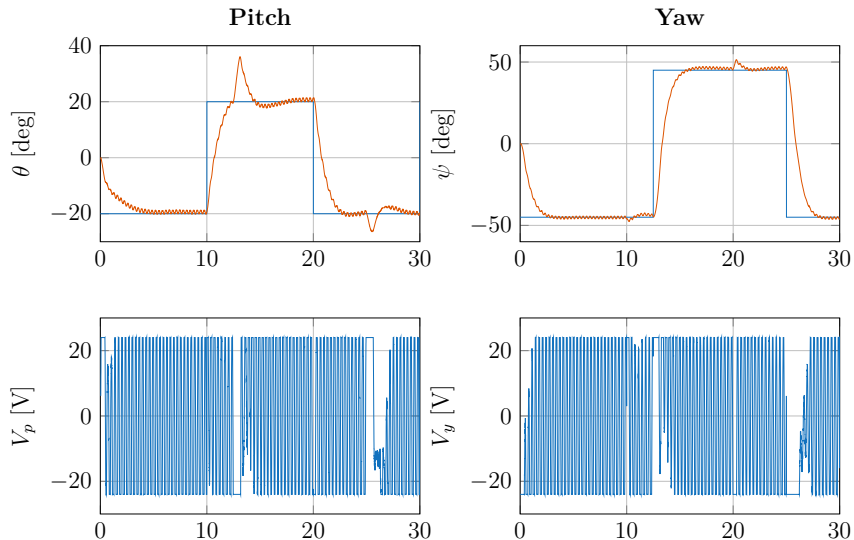
Problem: high inputs forced into the constraints

Solution: controller that considers the constraints in the calculation

LQI controller 2/3 - Simulator results



LQI controller 3/3 - Real plant results



MPC target setpoint 1/3 - approach and remarks

Approach

- discretization method: `zoh`

$$\dot{x}(t) \approx \frac{x(t + T_s) - x(t)}{T_s}$$

- the sampling time: $T_s = 0.2 \text{ s}$
- $Q = \text{diag}(10^4 \quad 5 \cdot 10^3 \quad 10 \quad 50)$
- $R = 0.1 I_2$, $P = 10Q$
- $N_{pred} = 20$ samples
- no constraints on the Δu
- setpoint: current value of the reference $\Rightarrow$ preemptive behaviour
- CasADi solver: `qrqp`
- Simulink implementation: `S Function` block:
 - generated c code from the CasADi script

Remarks

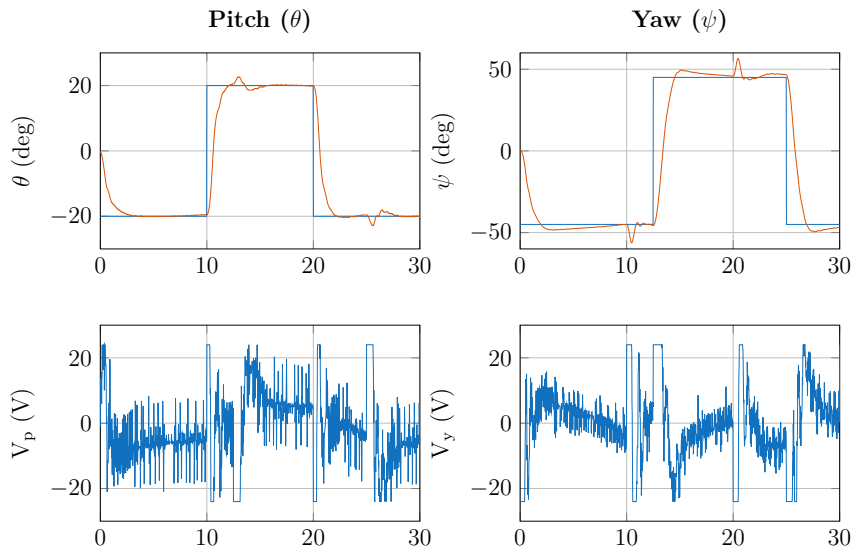
- lower command inputs
- $\epsilon = 0\%$
- high sensitivity, marked by a big σ'

Next step: MPC reference tracking

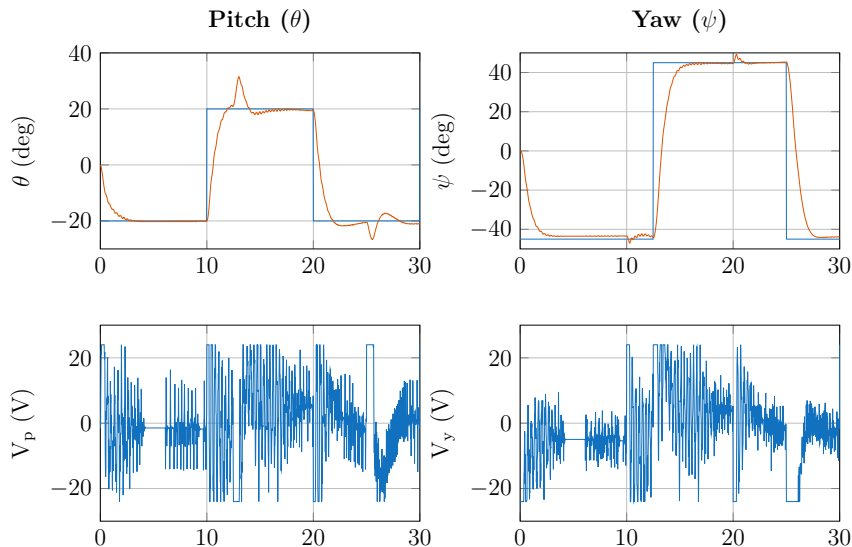
Problem: high sensitivity caused by the lack of preemptive behaviour

Solution: controller with preemptive behaviour

MPC target setpoint 2/3 - Simulator results



MPC target setpoint 3/3 - Real plant results



MPC reference tracking 1/3 - simulation

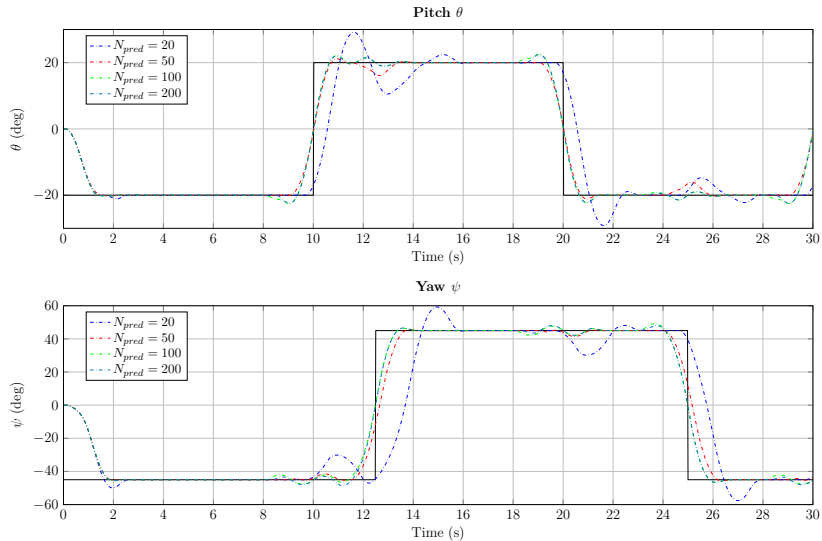
Approach

- discretization method: `zoh`
- the sampling time: $T_s = 0.02 \text{ s}$
- $Q = \text{diag} \begin{pmatrix} 200 & 0.2 & 10 & 1 \end{pmatrix}$
- $R = 0.002 I_2$, $P = 100Q$
- $-5 \leq \Delta u_i \leq +5$
- MATLAB simulation with the discrete linear model
- CasADi solver: `ipopt`
- varied N_{pred}

Remarks

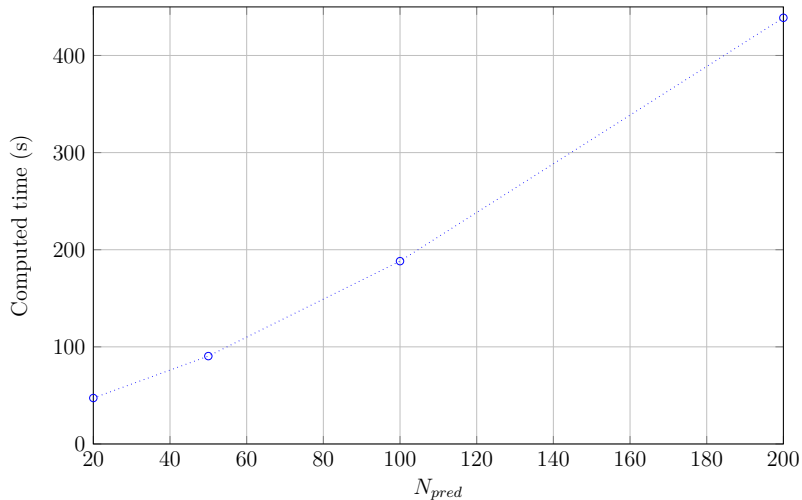
- *preemptive behaviour*
- bigger $N_{pred} \implies$ smaller overshoots, but more oscillations
- bigger $N_{pred} \implies$ bigger computational time

MPC reference tracking 2/3 - output results



MPC reference tracking 3/3 - computational time vs N_{pred}

Online computation vs prediction horizon



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Conclusions. Future work

Conclusions

- *Hypothesis confirmed*
 - *the real plant needs bigger inputs than the simulator for similar results*
- classical controllers, while simple, have numerous problems in the practical applications of the plant:
 - no physical constraint handling
 - no preemptive behaviour
 - commands with high amplitudes and high frequencies
- the predictive methods, while having complexity and convergence issues, are practical solutions to the problems of the classical methods

Future work

- Simulink implementation of the MPC controller with tracking reference
- embedded implementation of the LQR (for comparison) and the predictive control methods
- implementation of the Data-Driven Predictive Control

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